Run report: H6-large-run-20261018T2210Z

Descriptive record of one recorded run of examples/n2_routes.rs, mode run at 44 and 48 bits (Addendum H6), on a RunPod CPU pod under `research/weil_descent_angles_20261018/PROTOCOL.md` (Addendum A and Amendments B–G and Addendum H, each committed before the run it governs). It lists what ran, on what host, and what the tool recorded, file by file. Interpretation is in the study's `REPORT.md`. Costs are counted operations (F_p multiplications and inversions, group operations, native operations), never wall-clock; the only times are the tool's own seconds. Isolation: none earned (a shared cloud container, or a RunPod CPU pod whose lscpu is in host.txt); this affects only the seconds, since every recorded value is an exact count or a counted operation total.

env.txt

```
commit 6083c58aa45e04e4d3194689b3b3e6ef5c7be115 (fetched by the pod from raw.githubusercontent.com, sha256-checked)
cmd: rustc --edition 2021 -O examples/n2_routes.rs -o n2 && N2_THREADS=16 N2_SIZES=44,48 N2_CS=2 ./n2 "$OUT" run
host: RunPod CPU pod vv1r308vkkoewx (h6-large-sizes), 16 vCPU, flavors cpu5c,cpu3c, image rust:1-bookworm, launched 2026-10-18T22:09:01Z
cpu: AMD EPYC 4564P 16-Core Processor
rustc 1.99.1 (58cf8f9c3 2026-10-13)
runner: research/weil_descent_angles_20261018/runpod/podrun.py
```

tool_log.txt

```
jobs 6 total 514.8s
```

host.txt

```
Architecture:                         x86_64
CPU op-mode(s):                       32-bit, 64-bit
Address sizes:                        48 bits physical, 48 bits virtual
Byte Order:                           Little Endian
CPU(s):                               32
On-line CPU(s) list:                  0-31
Vendor ID:                            AuthenticAMD
Model name:                           AMD EPYC 4564P 16-Core Processor
CPU family:                           25
Model:                                97
Thread(s) per core:                   2
Core(s) per socket:                   16
Socket(s):                            1
Stepping:                             2
CPU(s) scaling MHz:                   33%
CPU max MHz:                          5881.0000
CPU min MHz:                          545.0000
BogoMIPS:                             8983.84
Flags:                                fpu vme de pse tsc msr pae mce cx8 apic sep mtrr pge mca cmov pat pse36 clflush mmx fxsr sse sse2 ht syscall nx mmxext fxsr_opt pdpe1gb rdtscp lm
constant_tsc rep_good amd_lbr_v2 nopl nonstop_tsc cpuid extd_apicid aperfmperf rapl pni pclmulqdq monitor ssse3 fma cx16 sse4_1 sse4_2 movbe popcnt aes xsave avx f16c rdrand lahf_lm cmp_legacy
svm extapic cr8_legacy abm sse4a misalignsse 3dnowprefetch osvw ibs skinit wdt tce topoext perfctr_core perfctr_nb bpext perfctr_llc mwaitx cpb cat_l3 cdp_l3 hw_pstate ssbd mba perfmon_v2 ibrs
ibpb stibp ibrs_enhanced vmmcall fsgsbase bmi1 avx2 smep bmi2 erms invpcid cqm rdt_a avx512f avx512dq rdseed adx smap avx512ifma clflushopt clwb avx512cd sha_ni avx512bw avx512vl xsaveopt xsavec
xgetbv1 xsaves cqm_llc cqm_occup_llc cqm_mbm_total cqm_mbm_local user_shstk avx512_bf16 clzero irperf xsaveerptr rdpru wbnoinvd cppc arat npt lbrv svm_lock nrip_save tsc_scale vmcb_clean
flushbyasid decodeassists pausefilter pfthreshold avic v_vmsave_vmload vgif x2avic v_spec_ctrl vnmi avx512vbmi umip pku ospke avx512_vbmi2 gfni vaes vpclmulqdq avx512_vnni avx512_bitalg
avx512_vpopcntdq rdpid overflow_recov succor smca fsrm flush_l1d
Virtualization:                       AMD-V
L1d cache:                            512 KiB (16 instances)
L1i cache:                            512 KiB (16 instances)
L2 cache:                             16 MiB (16 instances)
L3 cache:                             64 MiB (2 instances)
NUMA node(s):                         1
NUMA node0 CPU(s):                    0-31
Vulnerability Gather data sampling:   Not affected
Vulnerability Itlb multihit:          Not affected
Vulnerability L1tf:                   Not affected
Vulnerability Mds:                    Not affected
Vulnerability Meltdown:               Not affected
Vulnerability Mmio stale data:        Not affected
Vulnerability Reg file data sampling: Not affected
Vulnerability Retbleed:               Not affected
Vulnerability Spec rstack overflow:   Mitigation; Safe RET
Vulnerability Spec store bypass:      Mitigation; Speculative Store Bypass disabled via prctl
Vulnerability Spectre v1:             Mitigation; usercopy/swapgs barriers and __user pointer sanitization
```

```
Vulnerability Spectre v2:             Mitigation; Enhanced / Automatic IBRS; IBPB conditional; STIBP always-on; RSB filling; PBRSB-eIBRS Not affected; BHI Not affected
Vulnerability Srbds:                  Not affected
Vulnerability Tsx async abort:        Not affected
               total        used        free      shared  buff/cache   available
Mem:             124          14          69           0          42         110
Swap:              0           0           0
16
Linux 526a2cc5ccb1 6.8.0-51-generic #52~22.04.1-Ubuntu SMP PREEMPT_DYNAMIC Mon Dec  9 15:00:52 UTC 2 x86_64 GNU/Linux
rustc 1.99.1 (58cf8f9c3 2026-10-13)
```

rusage.txt

```
Command being timed: "bash -c mkdir -p examples && curl -sSfL https://raw.githubusercontent.com/aburan28/crypto/6083c58aa45e04e4d3194689b3b3e6ef5c7be115/examples/n2_routes.rs -o examples/
n2_routes.rs && echo 'fb19b56a3ed276e163c2cbc84f496014983e6f2b4376e2ea090a8423009afc7a  examples/n2_routes.rs' | sha256sum -c - && rustc --edition 2021 -O examples/n2_routes.rs -o n2 &&
N2_THREADS=16 N2_SIZES=44,48 N2_CS=2 ./n2 "$OUT" run"
  User time (seconds): 1610.12
  System time (seconds): 22.75
  Percent of CPU this job got: 315%
  Elapsed (wall clock) time (h:mm:ss or m:ss): 8:37.15
  Average shared text size (kbytes): 0
  Average unshared data size (kbytes): 0
  Average stack size (kbytes): 0
  Average total size (kbytes): 0
  Maximum resident set size (kbytes): 3342800
  Average resident set size (kbytes): 0
  Major (requiring I/O) page faults: 44
  Minor (reclaiming a frame) page faults: 15953532
  Voluntary context switches: 1251
  Involuntary context switches: 4870
  Swaps: 0
  File system inputs: 0
  File system outputs: 12456
  Socket messages sent: 0
  Socket messages received: 0
  Signals delivered: 0
  Page size (bytes): 4096
  Exit status: 0
```

summary.txt

```
O/J bits 44 arm J c 2: IC/rho ratio in [0.988, 2.961] over 12 cells -> unresolved; steps/p 0.295 (rho 0.328)
O/J bits 44 arm D c 2: IC/rho ratio in [28.403, 80.874] over 6 cells -> rho cheaper in every cell; steps/p 0.342 (rho 0.328)
O/J bits 48 arm J c 2: IC/rho ratio in [0.768, 2.316] over 12 cells -> unresolved; steps/p 0.311 (rho 0.448)
O/J bits 48 arm D c 2: IC/rho ratio in [28.134, 86.838] over 6 cells -> rho cheaper in every cell; steps/p 0.433 (rho 0.448)
O/J scaling arm R c 0: cost ~ p^1.213
O/J scaling arm J c 2: cost ~ p^1.038
O/J scaling arm D c 2: cost ~ p^1.263
K6 bits 44: rho steps/p mean 0.3284 (s.e. 0.0262, 30 runs) vs 0.886*sqrt(ell)/p mean 0.3132
K6 bits 48: rho steps/p mean 0.4478 (s.e. 0.0418, 30 runs) vs 0.886*sqrt(ell)/p mean 0.4430
O/J all answers verified: true
```

ranking.csv (first 4 of 4 rows)

bits	arm	c	cells	ic_cheaper_cells	rho_cheaper_cells	min_ratio	max_ratio	verdict
44	J	2	12	2	10	0.9879	2.9613	unresolved
44	D	2	6	0	6	28.4034	80.8740	rho cheaper in every cell
48	J	2	12	5	7	0.7681	2.3163	unresolved
48	D	2	6	0	6	28.1337	86.8385	rho cheaper in every cell

routes.csv (first 12 of 180 rows)

bits	instance	p	N	h	ell	target	arm	c	factor_base	steps	relations	b_relations	jac_ops	walk_M	walk_I	walk_native	jac_M	jac_I	la_M	la_I	la_native	eadd_M	eadd_I	verified	seconds
44	0	4330457	18752853744904	8	2344106718113	0	R	0	0	2070002	0	0	0	25218491	2089538	4111831	0	0	0	0	0	12	1	1	0.267
44	0	4330457	18752853744904	8	2344106718113	0	J	2	4162	1717014	858442	12	1771393	79756715	1770564	5159128	72838803	1770513	0	0	6869824	12	1	1	0.925
44	0	4330457	18752853744904	8	2344106718113	0	D	2	4162	1682105	662172	10	0	1876790343	16714828	8535144	0	0	0	0	4742902	12	1	1	7.456
44	0	4330457	18752853744904	8	2344106718113	1	R	0	0	476301	0	0	0	5896036	486858	945786	0	0	0	0	0	12	1	1	0.063
44	0	4330457	18752853744904	8	2344106718113	1	J	2	4162	1914199	959517	25	1974158	88892022	1973241	5755835	81174187	1973181	0	0	8180299	12	1	1	1.073
44	0	4330457	18752853744904	8	2344106718113	1	D	2	4162	1454316	571581	5	0	1623598570	14451772	7375552	0	0	0	0	3894459	12	1	1	6.497
44	0	4330457	18752853744904	8	2344106718113	2	R	0	0	704078	0	0	0	8655340	715770	1398488	0	0	0	0	0	12	1	1	0.094
44	0	4330457	18752853744904	8	2344106718113	2	J	2	4162	1600517	800253	6	1651352	74356015	1650590	4809754	67903816	1650535	0	0	6190481	12	1	1	0.832
44	0	4330457	18752853744904	8	2344106718113	2	D	2	4162	401367	157967	0	0	448230106	3995327	2047208	0	0	0	0	860130	12	1	1	1.782
44	0	4330457	18752853744904	8	2344106718113	3	R	0	0	2214124	0	0	0	26967547	2234461	4397464	0	0	0	0	0	12	1	1	0.293
44	0	4330457	18752853744904	8	2344106718113	3	J	2	4162	678118	338643	1	702016	31637978	701690	2041865	28876297	701638	0	0	2033360	12	1	1	0.347
44	0	4330457	18752853744904	8	2344106718113	3	D	2	4162	1418200	557040	0	0	1582554912	14090291	7186926	0	0	0	0	3769093	12	1	1	6.276

routes_summary.csv (first 12 of 36 rows)

bits	arm	c	ipr	jac_pricing	native_w	runs	verified	mean_p	mean_cost	mean_rho_cost	ratio_to_rho	mean_steps_over_p	rho_steps_over_p
44	J	2	3M	measured	0	30	1	4423334	64895366.0	22031611.5	2.9456	0.2952	0.3284
44	J	2	3M	measured	0.1	30	1	4423334	65780251.7	22318725.2	2.9473	0.2952	0.3284
44	J	2	3M	measured	1	30	1	4423334	73744223.3	24902748.1	2.9613	0.2952	0.3284
44	J	2	3M	eadd	0	30	1	4423334	25551720.3	22031611.5	1.1598	0.2952	0.3284
44	J	2	3M	eadd	0.1	30	1	4423334	26436606.0	22318725.2	1.1845	0.2952	0.3284
44	J	2	3M	eadd	1	30	1	4423334	34400577.6	24902748.1	1.3814	0.2952	0.3284
44	J	2	2log2p_M	measured	0	30	1	4423334	120482153.7	82160142.4	1.4664	0.2952	0.3284
44	J	2	2log2p_M	measured	0.1	30	1	4423334	121367039.5	82447256.1	1.4721	0.2952	0.3284
44	J	2	2log2p_M	measured	1	30	1	4423334	129331011.0	85031279.0	1.5210	0.2952	0.3284
44	J	2	2log2p_M	eadd	0	30	1	4423334	81166587.3	82160142.4	0.9879	0.2952	0.3284
44	J	2	2log2p_M	eadd	0.1	30	1	4423334	82051473.0	82447256.1	0.9952	0.2952	0.3284
44	J	2	2log2p_M	eadd	1	30	1	4423334	90015444.6	85031279.0	1.0586	0.2952	0.3284

Graph

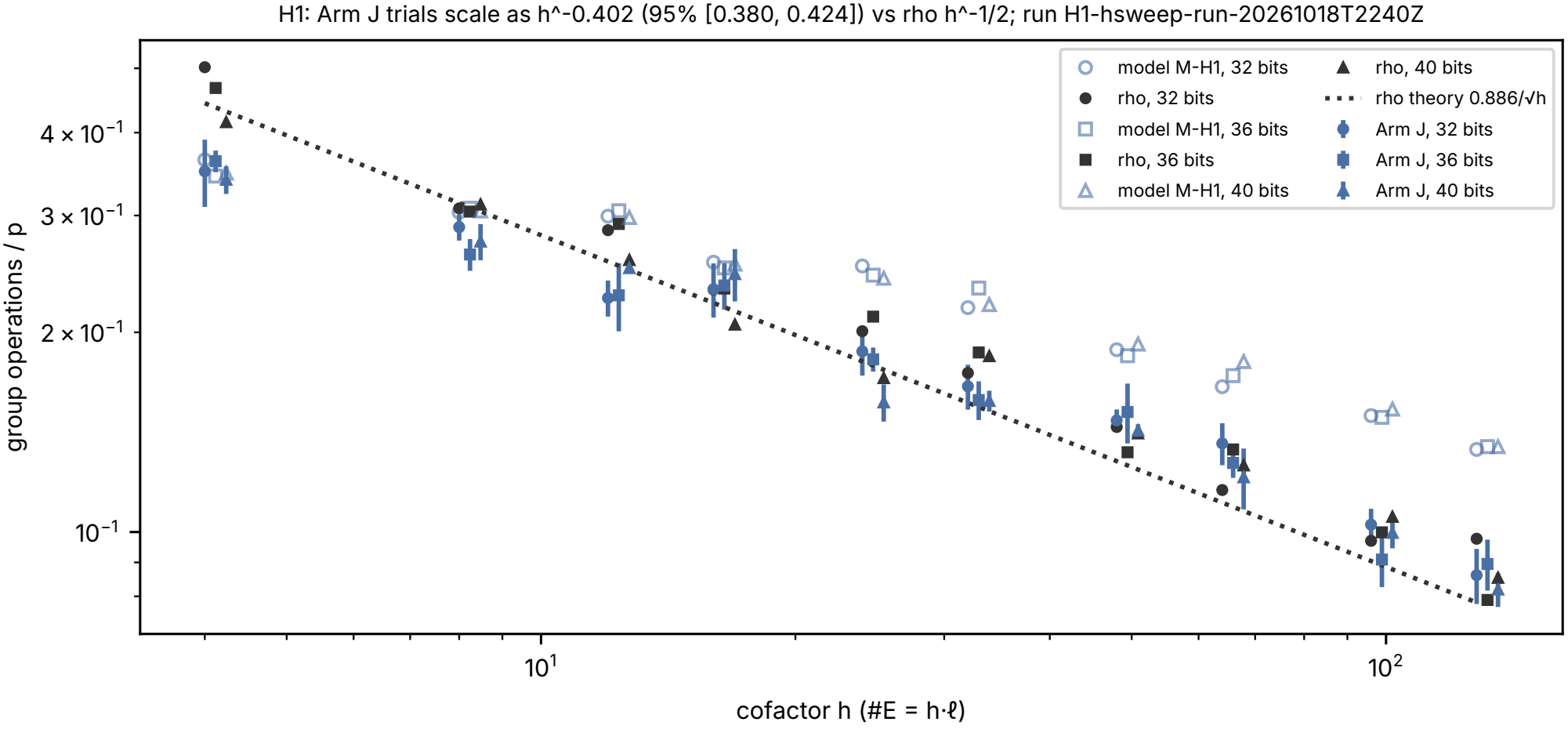


Figure 1: H6 values are in this report's summary; graph shown: H1 of the same round.